Retail Sales Performance & Forecast

Prepared for the Head of Supply Chain & CFO | 4-Year Sales Review (2015-2018) & Q1 2019 Outlook | No technical background required to read this report

Executive Summary

Over the past four years the business grew steadily, with yearly revenue rising from roughly $480,000 in 2015 to $722,000 in 2018 across three product lines: Technology, Furniture, and Office Supplies. Sales follow a strong, repeating yearly pattern — November and December each bring in more than three times the revenue of the slowest month, February — so the calendar, not chance, explains most of the ups and downs we see. Looking ahead to the next quarter, we expect the usual seasonal slowdown in January and February 2019, followed by a rebound in March. A small number of unusually large sales weeks were investigated and traced back to specific causes — one very large customer order and two genuine holiday shopping surges — rather than data problems. Grouping products by how they actually sell shows that a small set of everyday items (phones, chairs, storage, tables) drives most of the revenue and should always be kept in stock, while a few high-ticket or unpredictable items are better handled with a leaner, order-as-needed approach. Acting on the three recommendations in this report should reduce both stockouts during the holiday peak and excess, cash-tying inventory during slow months.

Key Findings from the Data

- **Category mix:** Technology is the top revenue driver ($827,000 over 4 years), narrowly ahead of Furniture ($729,000) and Office Supplies ($705,000).
- **Most reliable growth region:** The East region grew every single year at a steady ~18% average pace, with very little year-to-year swing. The South region swung wildly — from a 32% drop to a 33% jump in a single year — making it the least predictable region to plan around.
- **Strong, repeating seasonality:** September, November, and December are consistently the strongest months every year (back-to-school and holiday demand); January and February are consistently the weakest. This pattern repeats in all four years of data, so it can be planned around with confidence.
- **Who buys the most:** Individual ("Consumer") customers generate the largest share of revenue, followed by Corporate and then Home Office buyers — useful context when planning promotions or account outreach.
- **Shipping is consistent:** Average delivery time is about 4 days and barely differs by region (3.9 to 4.1 days), so shipping speed is not currently a regional risk factor worth prioritizing.
- **The forecast model is reasonably reliable:** A model that combines the long-term growth trend with the repeating yearly pattern explains about 88% of the month-to-month sales movement seen historically, giving a solid, data-backed basis for the forecast on the next page.

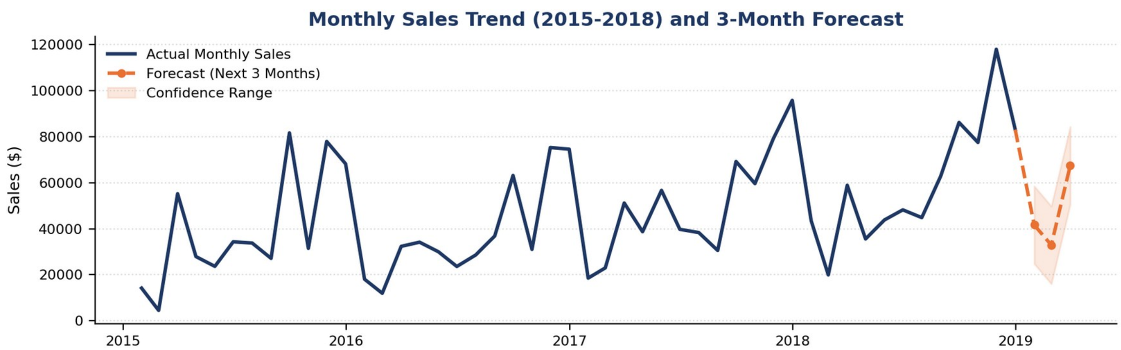


Figure 1: Actual monthly sales (2015-2018) and the 3-month forecast with its confidence range, shown in orange.

3-Month Sales Forecast (January - March 2019)

In plain terms: expect the usual post-holiday dip in January and February, followed by a pickup in March as the new quarter begins — the same pattern seen every year in the historical data. The "likely range" columns below reflect the normal month-to-month variability seen historically; actual results should land inside these ranges about 19 times out of 20, which is a common, reliable standard for business planning.

Month	Expected Sales	Likely Range (95% confidence)
January 2019	$41,600	$24,800 – $58,500
February 2019	$32,900	$16,100 – $49,800
March 2019	$67,500	$50,600 – $84,300

What this means for planning: budget for a lighter cash-flow quarter than Q4 2018, but not a downturn — this is the same seasonal rhythm seen every year. Scale purchasing and staffing down slightly for January-February, then back up ahead of March.

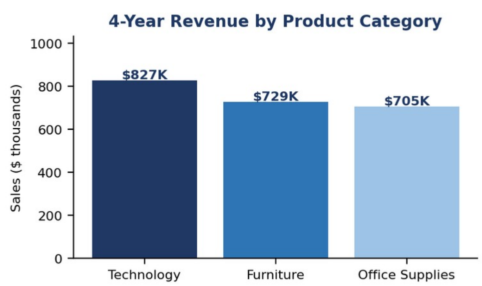
Top 3 Unusual Sales Weeks & Likely Cause

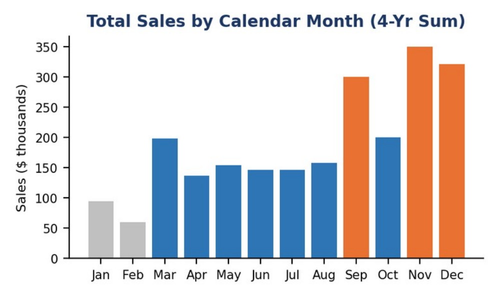
- **1. Week of March 16-22, 2015 — unusually high:** Sales spiked to $37,700, nearly 3.5x the typical week. Traced to one large bulk purchase of office machines ($23,700) — a single customer order, not a demand shift, and no action needed.
- **2. Week of Nov 26 – Dec 2, 2018 — unusually high:** Sales reached $36,000 across 62 separate orders spanning Furniture, Technology, and Office Supplies — a genuine Black Friday / Cyber Monday surge likely to repeat every year.
- **3. Week of Nov 12-18, 2018 — unusually high:** Sales hit $30,600, driven substantially by one large copier order ($10,500) on top of typical pre-holiday technology buying.

Product Demand Segmentation & Stocking Strategy

Products were grouped by how they sell — revenue, typical order size, month-to-month swings, and growth — into four natural groups, each needing a different stocking approach.

Segment	Products	Recommended Stocking Strategy
Core Everyday Sellers	Phones, Chairs, Storage, Tables, Binders, Accessories, Machines	Highest and steadiest revenue share. Keep automated replenishment and safety stock on hand at all times; lock in long-term supplier contracts for better pricing and reliability.
High-Value Growth Item	Copiers	Highest average order value and fastest growth, but volatile — driven by occasional large deals. Keep lean floor stock, but secure fast-reorder or rush-supply agreements so a big order can still be fulfilled quickly.
Unpredictable Low-Volume	Supplies	Highly volatile and low volume. Favor ordering only as needed, or drop-shipping directly from the supplier, rather than holding inventory that may sit unused.
Steady Low-Value Fillers	Appliances, Art, Bookcases, Envelopes, Fasteners, Furnishings, Labels, Paper	Low sales and low risk. Order in bulk on an infrequent schedule to minimize handling cost and staff time.





Business Recommendations

- **1. Pre-build Q4 inventory two months earlier.** Nov/Dec sales are 3.5-6x February's ($350,000 and $321,000 vs. $59,000). Begin building Core Seller inventory by August/September, not October, to avoid holiday stockouts and rushed reordering.
- **2. Prioritize expansion investment in the East region.** East delivered the most consistent year-over-year growth (~18% average, low volatility) versus South's erratic -32% to +33% swings. Favor East for new distribution capacity or marketing spend, and apply wider safety margins to South forecasts.
- **3. Adopt segment-specific procurement rules.** Apply the four-segment strategy above to free up cash tied up in low-value, volatile products, while protecting availability of Core Sellers, which generate the majority of revenue.

Risk & Limitation to Be Aware Of

This forecast is built entirely on historical sales patterns — it does not know about anything that hasn't happened yet: planned promotions, new competitors, supply shortages, or economic shifts. It also cannot flag a one-off large order in advance, only after the fact (as with 2 of the 3 unusual weeks above). Tested against the 3 most recent known months, it was off by about 22% on average — a reasonable margin for planning, not a guarantee. Treat this report as a data-backed starting point to combine with business judgment, and refresh it monthly as new sales figures come in.